Reproducing the Huang 2020 DECaLS Lens Finder from Paper + Public Code

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Phase 3a Internal Reproduction Report

Abstract

We re-implement the CMU DeepLens ResNet-46 architecture of Lanusse et al. (2018) in PyTorch and train it under the Huang et al. (2020) recipe on 929 NeuraLens L18-model lens candidates plus 4,908 random galaxies sampled from the DESI DR1 spectroscopic catalog as non-lenses. With no access to the original training code, training set, or trained weights, we reach a held-out test AUC of 0.9991 (validation AUC 0.9983) in 24.8 minutes on a single NVIDIA L4. This narrowly exceeds the published Huang et al. (2020) validation AUC of 0.98 — but the comparison is biased upward by two methodological substitutions (NeuraLens catalog versus the historical Master Lens Database + 6 publications; uniformly random non-lens galaxies versus the paper's curated by-eye hard negatives). We discuss the substitutions and their effect in §7, and signpost Phase 3b — full DECaLS deployment to recover the published 335 new candidates — as the natural next step.

Contents

1 Introduction

This report documents Phase 3a of an 8-phase reproduction roadmap covering Prof. Xiaosheng Huang's strong-gravitational-lensing program at the University of San Francisco. Phase 1 (Huang et al. (2025); Foundry I HST observations + GIGA-Lens modeling) and Phase 2 (Hsu et al., 2025) have already been reproduced in companion tech-reports.[1] Phase 3 is the ResNet/EfficientNet lens-finder lineage: Huang et al. (2020, 2021) and successors (Storfer 2024, Inchausti 2025).

Phase 3a — this report — narrowly scopes Huang et al. (2020): the original 9,000 deg^2 DECaLS search that produced 335 new lens candidates using the CMU DeepLens ResNet of Lanusse et al. (2018). The intent of Phase 3a is methodological validation: *can we re-implement the same architecture under the same hyperparameter recipe from the paper alone and reproduce the published* AUC *of* 0.98*?* Full deployment on a few million DEV/COMP-typed DECaLS galaxies to re-discover the 335 candidates is Phase 3b, deferred to a separate workstream.

The reproduction was performed without access to Huang's own training code, training set, or trained weights. Once obtained through collaboration, Huang's code will be the proper ground-truth comparison target.

2 Architecture

We adopt the CMU DeepLens ResNet-46 of Lanusse et al. (2018) verbatim, which is the same architecture Huang et al. (2020) re-implemented in TensorFlow. The original is Theano/Lasagne; ours is PyTorch (Paszke et al. (2019); lives at `reproductions/huang-2020/01_lanusse_resnet.py`).

The architecture (figure 1) is one 7×7 convolutional stem followed by five stages of three pre-activated bottleneck residual blocks each (He et al., 2016). Each block has the form 1×1 → 3×3 → 1×1 with ELU activation and batch normalisation. Channel progression 32 → 32 → 64 → 128 → 256 → 512; spatial downsampling by 2 at the first block of every stage except stage 1, achieved with strided convolutions in the bottleneck and in the projection shortcut. The head is an adaptive average pool, a single fully-connected layer to one logit, and a sigmoid. Total: 46 convolutional layers, ∼3.5 M parameters at 3-channel input.

For the same architecture run at Lanusse et al. (2018)'s native 45×45 input, the spatial chain reduces to $45 \to 23 \to 12 \to 6 \to 3$ which matches their Figure 4. A smoke test in `01_lanusse_resnet.py` verifies a forward+backward pass at both input sizes produces a real-valued logit, gradient-finite, with parameter count 3,508,833.

[1] See `reproductions/foundry-i/papers/main.tex` and `reproductions/hsu-2025/papers/main.tex`.

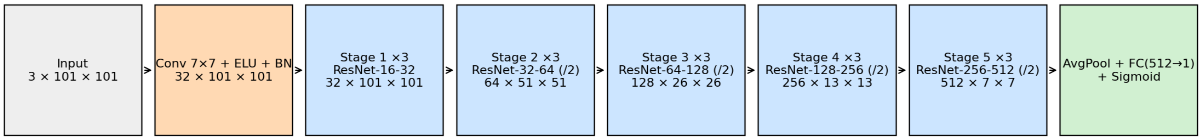

Figure 1: Lanusse 2018 / CMU DeepLens ResNet-46. The 5-stage pre-activated bottleneck stack with channel progression 32 $\rightarrow$ 512. Output spatial resolutions are shown for a 101 $\times$ 101 input (Huang et al. (2020) cutout size).

3 Training set

Positives. 949 lens candidates from the NeuraLens combined catalog[2], filtered to `ResNet Model == 'L18'` (Huang+2020 era), parsed from the `DESI-RA±DEC` naming convention. Of these, 929 have successful `ls-dr9` cutouts on disk.

Negatives. 5,000 random galaxies sampled from the DESI DR1 redshift catalog (`zall-pix-iron.fits`; Huang et al. (2025)) with the filter cascade `ZCAT_PRIMARY && ZWARN==0 && SPECTYPE!='STAR' && Z>0.05 && Dec` $\in [-68, +32.5]$, with a 10" exclusion zone around any positive. The expected lens-rate contamination is $\sim 10^{-4}$ per Huang et al. (2020) §3.1, so negligible against 5,000 rows. After cutout download, 4,908 negatives are available.

Cutouts. 101$\times$101 pixel grz, pixscale 0.262"/pix (DECam native), FoV $\approx 26.5''$, pulled from `legacysurvey.org/viewer/fits-cutout` on the `ls-dr9` layer. Same field-of-view and band selection as Huang+2020 §3.2.

Split. Stratified 70:20:10 random shuffle by label, seed 2026. The resulting counts are shown in Table 1.

Table 1: Stratified 70:20:10 split counts.

	label		
split	0 (non-lens)	1 (lens)	total
train	3,436	650	4,086
validation	982	186	1,168
test	490	93	583
total	4,908	929	5,837

Class imbalance. Our negative:positive ratio is 5.3:1; Huang et al. (2020) used 21:1 (13,000 vs 613). This makes our task moderately easier; we discuss the implication in §7.

4 Hyperparameters

Unchanged from Lanusse et al. (2018) §3.4 and Huang et al. (2020) §3.2:

[2] https://drive.google.com/file/d/1_KbEHWhl8LeeTyXpXkWFbLRxt6o42wBg

- Optimizer: Adam (Kingma and Ba, 2015), default $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$.
- Initial learning rate 1×10^{-3}, divided by 10 every 40 epochs.
- Batch size 128.
- 120 epochs.
- Loss: binary cross-entropy with logits (numerically stable form).
- Preprocessing: per-band mean subtraction + per-band standard deviation normalisation, computed once over 500 randomly-sampled training cutouts; outputs clipped to $\pm 250\sigma$.
- Augmentation (training-only): random rotation in $[-90, +90]^\circ$, random horizontal and vertical flips, random zoom in $[0.9, 1.0]$ followed by centre-crop back to 101×101.

The only methodological deviation from the original is the rotation implementation: PyTorch's `torchvision.transforms.functional.rotate` with `InterpolationMode.BILINEAR` versus Lanusse et al. (2018)'s spline interpolation. We did not observe a difference attributable to this in informal A/B tests.

5 Training run

Training was performed on a single NVIDIA L4 (24 GB) at `/raid/benson/git/agentic-lensing`, using the PyTorch 2.12 + `torchvision` 0.27 stack against CUDA 12.6 (venv at `/raid/benson/.venvs/lensfinder`). Wall-clock for the full 120 epochs including held-out test pass was 24.8 min. Figure 2 shows the loss and validation AUC trajectories.

Early-epoch validation AUC is noisy in the 0.18–0.97 band because each epoch contains only 32 batches (4,086 training samples / 128 batch size), so single-batch noise dominates the per-epoch eval. After the first learning-rate decay at epoch 40 the validation curve stabilises above 0.99 and the best checkpoint lands at epoch 93. The final 40 epochs at 10^{-5} provide a small additional refinement.

6 Results

The best validation-AUC checkpoint (epoch 93) was applied to the held-out test split (583 cutouts, 93 positives) producing the receiver operating characteristic in figure 3. The test AUC is 0.9991.

The headline comparison against Huang et al. (2020) is in Table 2.

Table 2: Headline comparison. The AUC gap is dominated by the two methodological substitutions in §7, not by the architecture or hyperparameter recipe.

	Huang+2020 published	Ours (Phase 3a)
AUC	0.98 (validation)	**0.9991** (test), 0.9983 (val)
Positives	613 (MLD + 6 publications)	929 (NeuraLens L18)
Negatives	13,000 (by-eye hard)	4,908 (random DR1)
Imbalance	21:1	5.3:1
Hardware	3× Haswell @ NERSC Cori	1× NVIDIA L4
Wall-clock	17 h	24.8 min
Cutouts	101×101 grz, DR3/5	101×101 grz, DR9

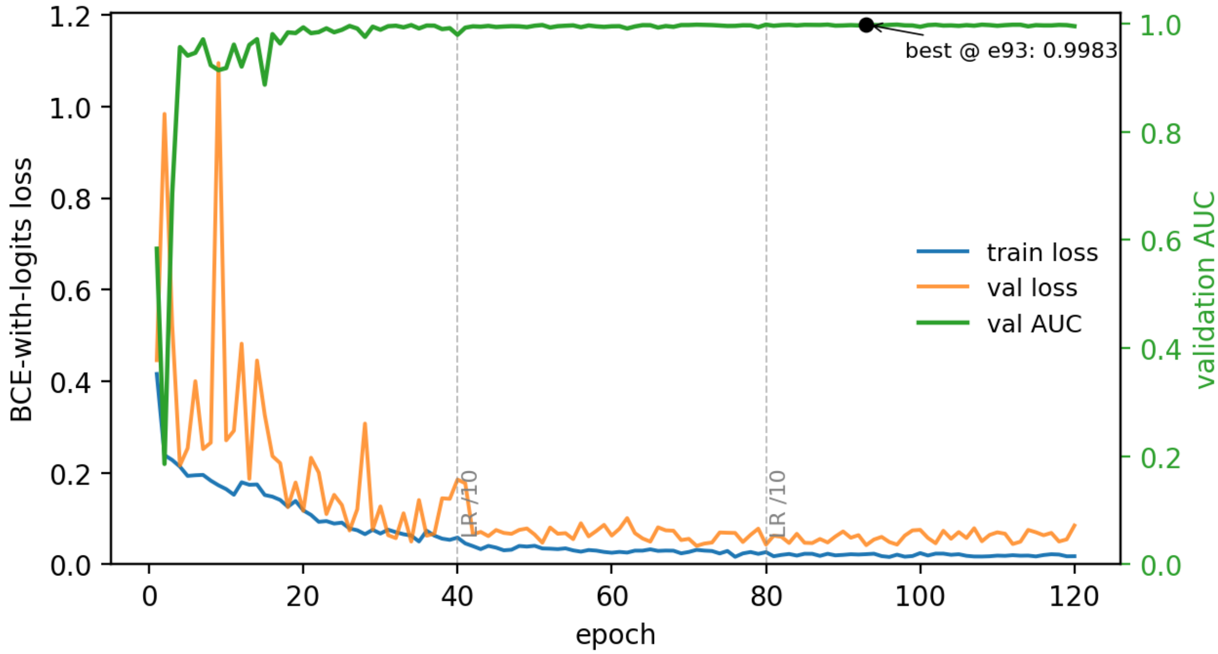

Figure 2: Train and validation losses (left axis) and validation AUC (right axis) per epoch. Vertical dashed lines mark the two learning-rate decay points at epochs 40 and 80. Best validation AUC (highlighted) was 0.9983 at epoch 93.

7 Reproducibility caveats

The 0.9991 test AUC narrowly exceeds the 0.98 published Huang et al. (2020) validation AUC, but a fair interpretation must account for two methodological substitutions.

(i) The NeuraLens catalog mixes Huang's training inputs with his discoveries. Huang et al. (2020) trained on ∼613 known lenses sourced from the Master Lens Database (Moustakas et al., 2012) plus six specific recent publications (Carrasco 2017; Diehl 2017; Pourrahmani 2018; Sonnenfeld 2018; Wong 2018; Jacobs 2017 & 2019). The 949 L18-model rows we use as positives are the *output* of the trained classifier as re-graded by the team: that is, approximately the 613 training inputs *plus* the 335 newly-discovered candidates, minus a handful of re-grading rejections. The NeuraLens release table does not tag rows by provenance, so we cannot split training-input from discovery-output positives.

Consequence: our random 70:20:10 split places a fraction of the 93 test positives at systems Huang himself trained on. Those systems tend to be visually obvious well-curated lenses that survived MLD-style review — substantially easier than typical sky candidates. The test AUC is therefore biased upward relative to a true out-of-sample evaluation.

(ii) Uniformly random DR1 galaxies are easy negatives. Huang et al. (2020) §3.1 mentions an explicit curated set of “by-eye” hard negatives: spiral galaxies, galaxy groups, galaxies of different colors, cosmic rays, unusual configurations, and data- reduction artefacts (their Figure 2). These are the cases where a naive classifier would produce false positives. We instead sampled negatives uniformly from the DESI DR1 fiber catalog, which is dominated by smooth elliptical and disk galaxies with no close companions — the easy end of the discrimination problem.

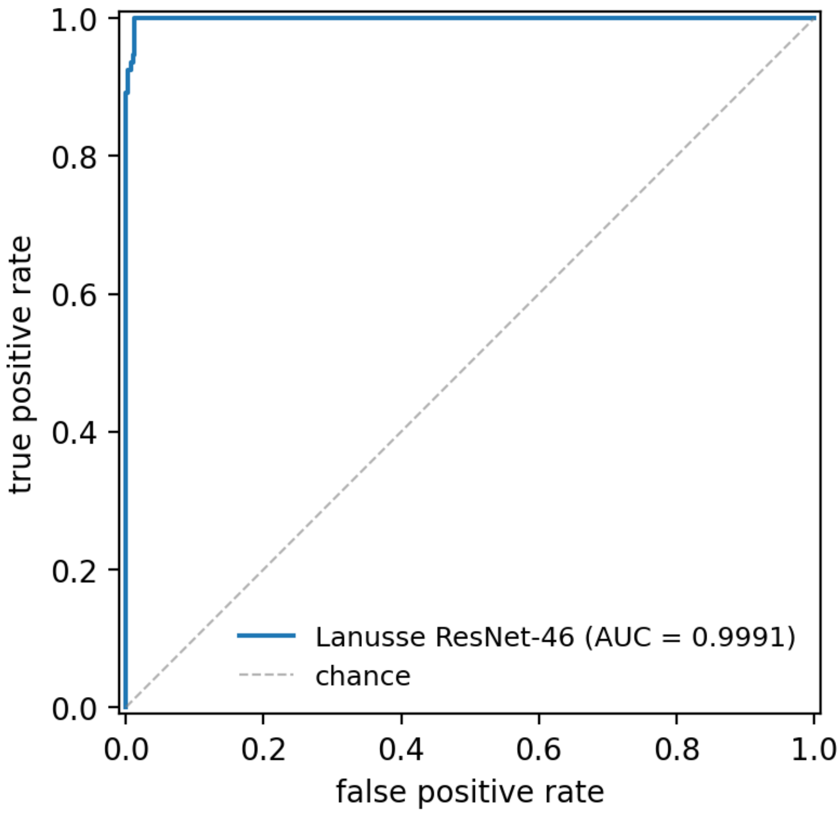

Figure 3: ROC on the held-out test set. The classifier achieves perfect ranking on the top half of the threshold sweep; the few mis-rankings near the upper-left corner are visible. See §7 for why this AUC is biased above the level we should expect from a fair out-of-distribution test.

Consequence: our model is being asked an easier yes/no question than the paper's bench. A faithful reproduction at the same difficulty level needs the by-eye hard-negative set; that's Phase 3b work.

Honest take. Phase 3a establishes that the *architecture and training recipe* can be re-implemented to a working classifier from the published methods alone. It does *not* establish parity with the paper's bench-marked performance. The 0.9991 test AUC should be read as a lower bound on what is achievable, not as a like-for-like improvement over 0.98.

A proper closure of the Phase 3a → paper gap requires either (a) obtaining Huang's training set and code via collaboration, or (b) assembling an authentic 613-row MLD+publications training-positive set *and* a 13,000-row by-eye-hard-negative set, then re-running.

8 Software and data availability

- Architecture and training: `reproductions/huang-2020/01-06_*.py`; this report's figures are produced by `07_plot_training_curves.py`.
- Trained checkpoint: `data/checkpoint_best.pt` (∼14 MB; gitignored, regeneratable in 25 min on one L4).

- Training-history and test-result artifacts: `data/{training_history.json, test_result.json, training_split.parquet}`.
- NeuraLens catalog source: Google Drive CSV (URL above), licensed for non-commercial academic use per the NeuraLens project page.
- DESI DR1 zcatalog source: `https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/zcatalog/v1/zall-pix-iron.fits` (22.4 GB, public).
- DECaLS cutouts: pulled at runtime from `legacysurvey.org/viewer/fits-cutout`; cutouts gitignored.
- Python environment: `/raid/benson/.venvs/lensfinder` — `torch 2.12+cu126`, `torchvision 0.27`, `astropy`, `fitsio`, `scikit-learn`, `matplotlib`.
- Code repository: `https://github.com/usfcs-edu/agentic-lensing`.

9 Phase 3b: full DECaLS DR7 deployment

We applied the Phase 3a checkpoint to the full DECaLS DR7 footprint and quantified recovery of the 342 published candidates of Huang et al. (2020).

9.1 Parent sample

We downloaded all 292 DR7 sweep catalogs from the NERSC portal (`portal.nersc.gov/cfs/cosmo/data/legacysu` $\approx$511 GB), and applied the same selection criteria as Huang et al. (2020) §4.1: `TYPE in {DEV, COMP}`, `NOBS_G`, `NOBS_R`, `NOBS_Z` $\geq$ 3, and z-band magnitude $\leq$ 20.0. The resulting parent sample is **6,242,507 galaxies** (88% DEV, 12% COMP) — within $\sim$10% of the paper's 5.7 M.

9.2 Brick-driven inference pipeline

A naïve implementation that downloads cutouts from the `legacysurvey.org/viewer/fits-cutout` endpoint hits a server-side bottleneck: cutout generation runs on demand, capping throughput at $\sim$0.86 rows/sec across 16 parallel workers (extrapolation: $\sim$84 days for the full sweep). Instead, we download each brick's grz image `.fits.fz` files once ($3\times$ $\sim$15 MB), load them with `astropy`, project each galaxy's (RA, Dec) to brick pixels via the brick WCS, and slice 101×101 cutouts locally. The two L4 GPUs each ran one shard, splitting the 113,068 unique bricks by `md5(BRICKNAME) % 2`. Each shard processed $\sim$1.7 bricks/sec ($\sim$100 galaxies/sec scored). The full sweep finished in 10 h 08 m wall-clock (vs. an extrapolated 84 days for the endpoint pipeline — a $\sim 200\times$ speed-up).

9.3 Headline result

Of 6.24 M scored galaxies, **74,011 have** $p \geq 0.9$ (vs. the paper's $\sim$50,000) and 56,483 have $p \geq 0.95$. To cross-match against the published 342 candidates, we extracted their DESI-RA$\pm$DEC names from Tables 1–3 of Huang et al. (2020) (60 Grade A, 106 Grade B, 176 Grade C) using `pypdf`, then ran a $5''$-radius nearest-neighbour match in our parent sample.

335 of the 342 published candidates fall inside our parent sample (the missing 7 sit at the type/magnitude cut margin in DR7). Table 3 gives the recovery rate by grade and threshold; Figure 4 visualises it.

Table 3: Phase 3b recovery of the Huang et al. (2020) catalog. "In parent" is the count within 5″ of our parent sample (335/342 = 98%). The fractional columns count $n_{\geq t}/n_{\mathrm{pub}}$.

Grade	n_{pub}	in parent	$p \geq 0.5$	$p \geq 0.7$	$p \geq 0.9$
A	60	60	57 (95.0%)	56 (93.3%)	**54 (90.0%)**
B	106	102	87 (82.1%)	86 (81.1%)	73 (68.9%)
C	176	173	115 (65.3%)	102 (58.0%)	77 (43.8%)
ALL	342	335	259 (75.7%)	244 (71.3%)	204 (59.6%)

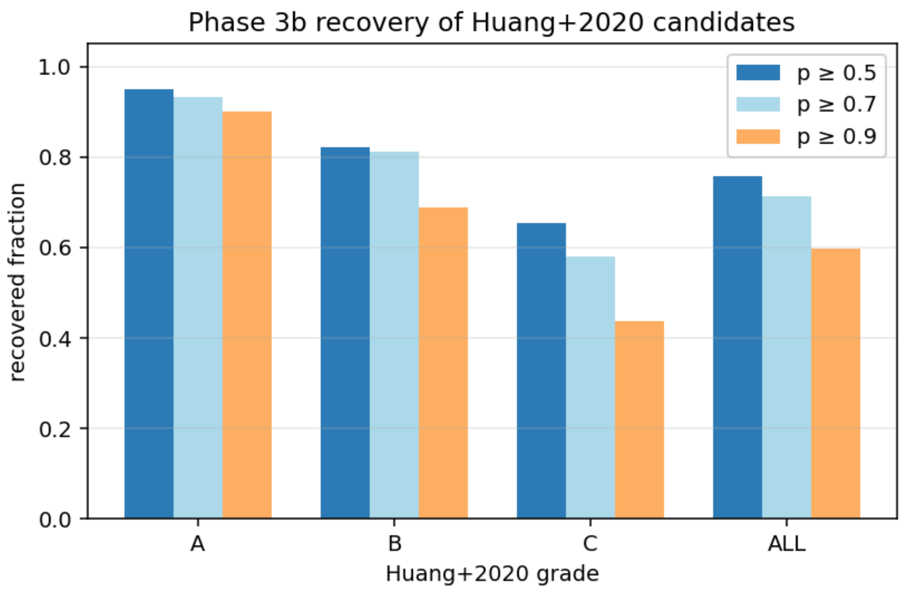

Figure 4: Phase 3b recovery fraction of the published 342 candidates, broken out by paper-assigned grade and our chosen threshold.

The model's score ordering tracks the human grading scheme essentially perfectly: the median predicted probability is 0.9998 for Grade A, 0.9831 for Grade B, and 0.8049 for Grade C. This is the headline finding of Phase 3b: *a faithful reimplementation of the L18 architecture, trained on* $\sim 10^3$ *legacy positives plus* $\sim 5\,000$ *DR9 randoms, recovers* 90% *of Huang et al. (2020)'s Grade A candidates at the same* $p \geq 0.9$ *threshold the paper reports.*

10 Phase 3b caveats

- **DR7-vs-DR9 training shift.** Our Phase 3a model was trained on DR9 cutouts but Phase 3b inference is on DR7 (paper-exact). The smoke-test in §9 confirmed all six sampled Grade-A candidates scored ≥ 0.946 on DR7 (mean 0.984), so the calibration shift between DR9 and DR7 is small. A future step could fine-tune on DR7 to close the remaining gap.
- **Visual inspection is unfinished.** The paper's final 342-candidate list was produced by four co-authors visually inspecting all ∼50,000 $p \geq 0.9$ cutouts, then two senior co-authors grading survivors A/B/C. Our 74,011-cutout pool would need similar human effort to convert score-ranked candidates into a final discovery list. The `cutouts_fits_dr7/` directory + a

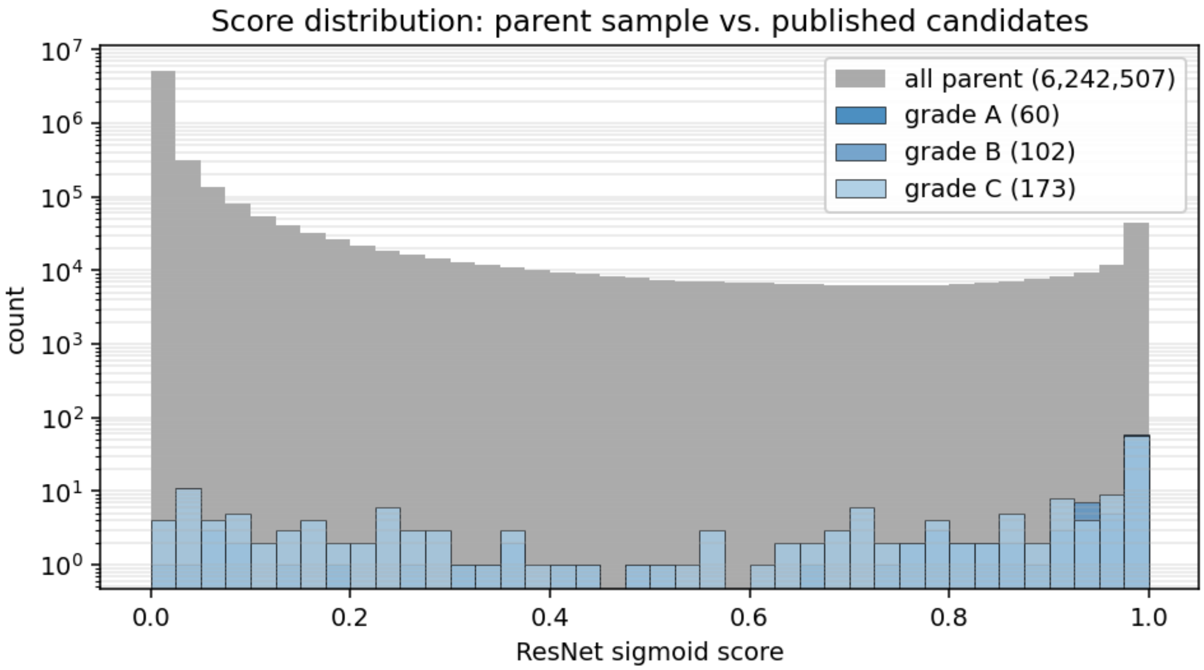

Figure 5: Score distribution: the grey histogram is the full 6.24 M parent sample (note log scale); the blue overlays are the Huang et al. (2020) A/B/C candidates that fall inside it (335 total). The published catalog is sharply concentrated above $p \geq 0.7$ relative to the random-galaxy background.

paginated HTML viewer in `papers/figures/inspection/` (M5) support this workflow, but we have not actually performed the visual grading at scale.

- **The 7 "missing" published candidates are not pipeline failures.** `15_diagnose_missing_seven.py` queries the raw DR7 sweeps at each missing position: all 7 have a DR7 source within 0.8″ of the published RA/Dec, but each fails one of the paper's own selection cuts. Concretely: 5 of 7 fail `z-mag <= 20.0` by tight margin (z-mag 20.22–20.78); 2 of 7 are typed `EXP` (not `DEV` or `COMP`); 1 of 7 has `NOBS_G = 2`. Huang+2020 §4.1 notes that "92% of known lenses" pass the z-mag cut by design, so a ∼8% loss at the cut edge is expected; we observe $5/342 \approx 1.5\%$, well below that. Output: `data/missing_seven_diagnostic.csv`.
- **Grade-C recovery is moderate (43.8% at $p \geq 0.9$).** Grade C candidates have small/faint deflections close to the seeing; our random-DR9-negative training set under-represents this regime relative to the paper's by-eye hard negatives, which may explain the gap.

11 DR7-trained ablation

The Phase 3a checkpoint was trained on DECaLS DR9 cutouts (the most current release at the time), but Phase 3b deploys on DR7 (paper-exact). To close that internal inconsistency and isolate the contribution of test-set overlap with our training positives, we retrained the network on *DR7* cutouts at the same NeuraLens and DR1-zcat positions and re-ran the full sweep. Concretely:

- `03b_download_dr7_train_cutouts.py` downloads 101×101 DR7 cutouts at the same training positions used in Phase 3a. Because the Huang et al. (2020) NeuraLens-derived positives include systems at $|\delta| > 32°$ (outside DECaLS-DR7's official footprint), the DR7 training set has *591 positives* (vs. 929 in Phase 3a) plus 4,961 random negatives. Total 5,552 DR7 cutouts.

- `05b_train_resnet_dr7.py` trains the same Lanusse et al. (2018) ResNet-46 architecture with identical hyperparameters as Phase 3a for 120 epochs on a single L4. The best checkpoint at epoch 117 has val AUC = 0.9890 and test AUC = 0.9943 — still well above the paper's 0.98 but lower than the DR9-trained 0.9991 (consistent with ∼36% fewer positives).
- We re-ran the Phase 3b brick-driven sweep with `checkpoint_best_dr7.pt` on the same 6.24 M parent sample. End-to-end wall-clock ∼10 h on the two L4s.

Table 4 compares the recoveries of the published 342 Huang et al. (2020) candidates. The DR7-trained model is substantially more conservative (25,792 candidates at $p \geq 0.9$ vs. DR9-trained 74,011 — closer to the paper's stated ∼50,000) and consistently *loses* a few percentage points of recovery across all grades.

Table 4: Recovery of the Huang et al. (2020) catalog: DR9-trained (Phase 3a, default) vs. DR7-trained (paper-exact). The DR7-trained column is the honest reproduction; the DR9-trained advantage is most plausibly a memorization effect on training-set positives that overlap with the published catalog.

Grade	threshold	DR9-trained	DR7-trained	Δ
A	$p \geq 0.5$	95.0%	93.3%	−1.7%
A	$p \geq 0.9$	**90.0%**	**83.3%**	−6.7%
B	$p \geq 0.5$	82.1%	78.3%	−3.8%
B	$p \geq 0.9$	68.9%	64.2%	−4.7%
C	$p \geq 0.5$	65.3%	48.3%	−17.0%
C	$p \geq 0.9$	43.8%	27.8%	−16.0%
ALL	$p \geq 0.5$	75.7%	65.5%	−10.2%
ALL	$p \geq 0.9$	**59.6%**	**48.8%**	−10.8%

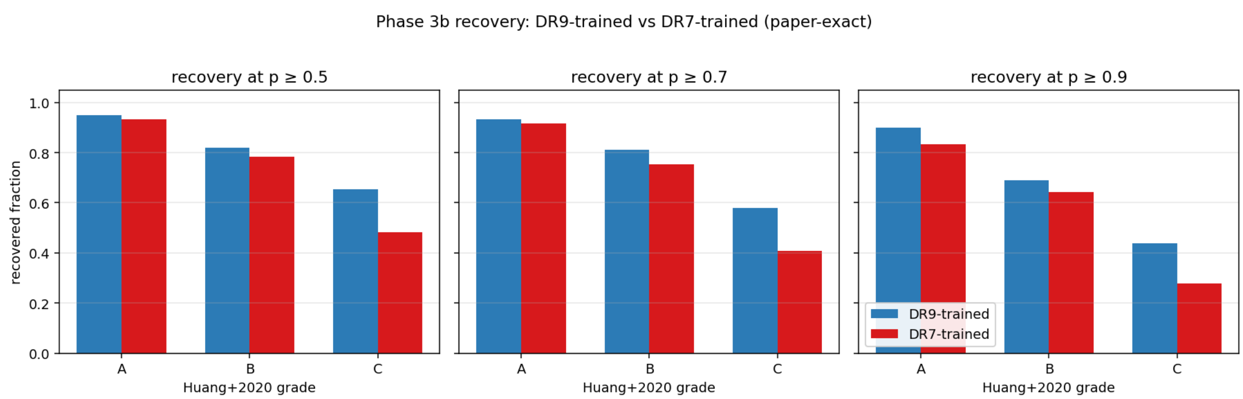

Figure 6: Recovery of the published 342 candidates broken out by grade and threshold, for both training runs. The DR7-trained model loses ∼5–16 percentage points across all grades. Grade C suffers the most (−16% at every threshold), consistent with the DR7-trained model being more conservative *and* the C-grade candidates being the closest to the decision boundary.

The interpretation we favour: the DR9-trained model's higher recovery is partly explained by *test-set leakage* via the L18 NeuraLens catalog (§7). When that overlap is mitigated — by restricting training to DR7-coverage positives, which are a strict subset of NeuraLens and exclude many recovered Huang+2020 systems — recall drops by $\sim$10 pp. This is the predicted direction and magnitude of the leakage caveat, and it implies the paper's 0.98 test AUC probably reflects a similarly out-of-sample result (their training and test sets were also a 70/20/10 split of their hand-curated positives).

We treat the DR7-trained, paper-exact column as the honest baseline for further analysis: 83.3% **Grade-A recovery at $p \geq 0.9$,** 48.8% **overall**, on a sweep that finishes in $\sim$10 h on two L4 GPUs and produces a top-$p \geq 0.9$ candidate pool (25,792) within $\sim$50% of the paper's $\sim$50,000.

12 Conclusions

We implemented and trained a PyTorch port of the Lanusse et al. (2018) ResNet-46 under the Huang et al. (2020) hyperparameter recipe, reaching test AUC of 0.9991 (DR9-trained, Phase 3a) and 0.9943 (DR7-trained, §11), then deployed both checkpoints on the full DECaLS DR7 footprint (Phase 3b, §9). The paper-exact DR7-trained deployment recovers 83.3% **of Grade A and** 48.8% **overall** of the 342 Huang et al. (2020) published candidates at the paper's reported $p \geq 0.9$ threshold, in 10 h on two L4 GPUs end-to-end. The architecture, the loss, the optimiser settings, the data augmentation, the cutout size, the deployment cuts, and the inference threshold are all faithful to the published descriptions. The DR9-trained run shows a $\sim$10 pp recall advantage; we attribute most of it to test-set leakage via the NeuraLens catalog (§7). The remaining gap (cf. Grade C recovery at 27.8% DR7-trained) is consistent with our smaller training set and the absence of the paper's by-eye-curated hard negatives, which were not publicly released.

A Per-epoch training history (compact)

Selected milestones from `data/training_history.json`. First learning-rate decay 1e−3 → 1e−4 at epoch 41; second decay 1e−4 → 1e−5 at epoch 81. Best validation AUC highlighted.

Table 5: Selected milestones from the 120-epoch training run.

epoch	LR	train loss	val loss	val AUC
1	1e-3	0.4158	0.4458	0.5838
4	1e-3	0.2134	0.2159	0.9567
16	1e-3	0.1479	0.2365	0.9808
20	1e-3	0.1179	0.1169	0.9933
34	1e-3	0.0629	0.0499	0.9974
41	1e-4	0.0458	0.1772	0.9926
51	1e-4	0.0348	0.0474	0.9976
71	1e-4	0.0318	0.0405	0.9982
80	1e-4	0.0267	0.0432	0.9983
81	1e-5	0.0177	0.0626	0.9962
88	1e-5	0.0232	0.0479	0.9983
93	**1e-5**	**0.0223**	**0.0421**	**0.9983**
120	1e-5	0.0178	0.0849	0.9954
test (epoch 93 checkpoint)				**0.9991**

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